

Requirement Analysis Document

Quotes Recommendation Bot using Rasa & NLP

1. Introduction

This document presents a detailed requirement analysis for a Quotes Recommendation Bot built using Rasa and Natural Language Processing (NLP). The system aims to provide personalized and context-aware quotes based on user mood, preferences, and interactions.

2. Project Objectives

- Deliver personalized motivational, inspirational, and thematic quotes.
- Understand user intent and emotional context using NLP.
- Provide a seamless conversational experience across platforms.
- Ensure scalability, performance, and maintainability.

3. Project Scope

In Scope:

- User interaction via chatbot interface.
- Quote recommendation based on mood, category, and history.
- NLP-based intent and entity recognition.

Out of Scope:

- Human counseling or psychological diagnosis.
- Voice-based interaction (initial phase).

4. Functional Requirements

- User can request quotes by mood or topic.
- Bot identifies user intent and entities.
- System recommends relevant quotes.
- User can like, dislike, or request another quote.
- Admin can manage and update quote database.

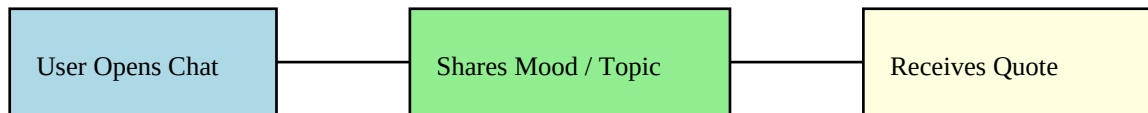
5. Non-Functional Requirements

- **Performance:** Response time under 2 seconds.
- **Scalability:** Support multiple concurrent users.
- **Security:** Secure APIs and data protection.
- **Usability:** Simple, intuitive conversational flow.
- **Availability:** 99% uptime.

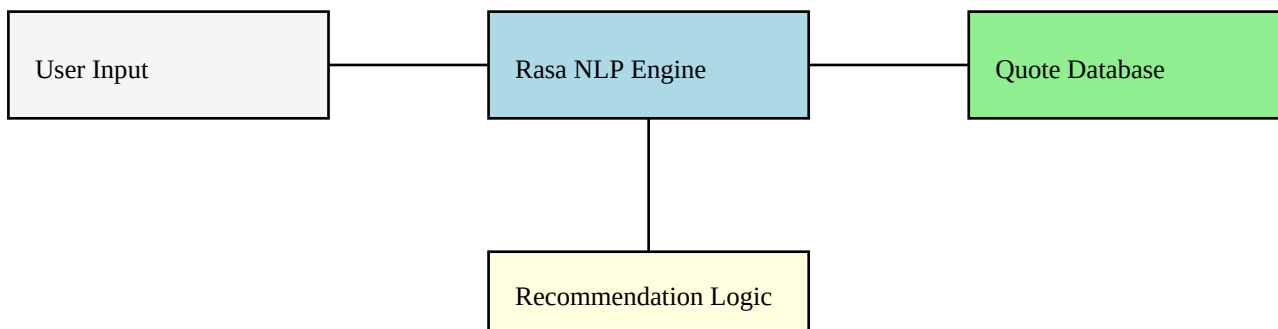
6. Constraints

- Limited training data for initial NLP models.
- Dependency on open-source frameworks.
- Budget and time constraints.

7. Customer Journey Map



8. Data Flow Diagram (DFD)



9. Solution Requirements

- NLP engine for intent classification and entity extraction.
- Recommendation logic for personalized quotes.
- Backend services for processing and data management.
- Frontend/chat interface integration.

10. Technology Stack

Layer	Technology
Chatbot Framework	Rasa
NLP	spaCy, TensorFlow
Backend	Python, FastAPI
Database	PostgreSQL / MongoDB
Deployment	Docker, AWS
Frontend	Web / Messaging Platforms

11. Conclusion

This requirement analysis provides a strong foundation for developing a scalable, intelligent, and user-centric Quotes Recommendation Bot. The proposed solution leverages modern NLP techniques and a robust technology stack to deliver meaningful user experiences.